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Batch: 15

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EXPERIMENT-18

1.Test Case

Test Case	Input	Expected Output	Type
TC1	ID=101, m=1, n=2; Course 1: code=1001, credit=4, grade=O; Course 2: code=1002, credit=3, grade=A	SGPA = 9.57, CGPA = 9.57	Valid
TC2	ID=102, m=2; Sem 1: n=2, (1001,3,A), (1002,4,B); Sem 2: n=2, (2001,3,O), (2002,2,C)	Sem 1 SGPA=8.43, Sem 2 SGPA=8.80, CGPA=8.58	Valid
TC3	ID entered as 0, then 103	Error for invalid ID; accepts 103 after re-entry	Invalid
TC4	m entered as -2, then 1	Error for invalid semesters; accepts 1 after re-entry	Invalid
TC5	n entered as 0, then 1	Error for invalid number of courses; accepts 1 after re-entry	Invalid
TC6	Course code entered as -10, then 1001	Error for invalid course code; accepts 1001	Invalid
TC7	Credit entered as 0, then 3	Error for invalid credit; accepts 3	Invalid
TC8	Grade entered as X, then B	Error for invalid grade; accepts B	Invalid

TC9	ID=1, m=1, n=1, code=1, credit=1, grade=P	SGPA=4.00, CGPA=4.00	Boundary
TC10	ID=1, m=1, n=1, code=1, credit=1, grade=O	SGPA=10.00, CGPA=10.00	Boundary

2. C Program

```
#include <stdio.h>

int main()
{
    int global_id, m, n;
    int semester, course;
    int code, credit, GP;
    int semester_TC, semester_TGC;
    int cumulative_TC, cumulative_TGC;
    char grade;
    float SGPA, CGPA;

    /* Input and validation of Global ID */
    do
    {
        printf("Enter Global ID: ");
        scanf("%d", &global_id);
        if (global_id <= 0)
            printf("Invalid Global ID. Please enter a positive value.\n");
    } while (global_id <= 0);

    /* Input and validation of number of semesters */
    do
    {
        printf("Enter number of semesters: ");
        scanf("%d", &m);
```

```

    if (m <= 0)

        printf("Invalid number of semesters. Please enter a positive value.\n");
} while (m <= 0);
cumulative_TC = 0;
cumulative_TGC = 0;
/* Loop for semesters */
for (semester = 1; semester <= m; semester++)
{
    semester_TC = 0;
    semester_TGC = 0;
    printf("\nSemester %d\n", semester);
    /* Input and validation of number of courses */
    do
    {
        printf("Enter number of courses: ");
        scanf("%d", &n);
        if (n <= 0)
            printf("Invalid number of courses. Please enter a positive value.\n");
    } while (n <= 0);
    /* Loop for courses */
    for (course = 1; course <= n; course++)
    {
        printf("\nCourse %d\n", course);
        /* Course code validation */
        do
        {
            printf("Enter course code: ");
            scanf("%d", &code);
            if (code <= 0)
                printf("Invalid course code. Please enter a positive value.\n");
        } while (code <= 0);
        /* Credit validation */

```

```

do
{
    printf("Enter course credit: ");
    scanf("%d", &credit);

    if (credit <= 0)
        printf("Invalid credit. Please enter a positive value.\n");
} while (credit <= 0);
/* Grade validation */
do
{
    printf("Enter grade (O/A/B/C/D/E/P): ");
    scanf(" %c", &grade);
    if (grade != 'O' && grade != 'A' &&
        grade != 'B' && grade != 'C' &&
        grade != 'D' && grade != 'E' &&
        grade != 'P')
    {
        printf("Invalid grade. Please enter O, A, B, C, D, E, or P.\n");
    }
} while (grade != 'O' && grade != 'A' &&
        grade != 'B' && grade != 'C' &&
        grade != 'D' && grade != 'E' &&
        grade != 'P');
/* Finding Grade Point */
switch (grade)
{
    case 'O':
        GP = 10;
        break;
    case 'A':
        GP = 9;

```

```

        break;
    case 'B':
        GP = 8;
        break;
    case 'C':
        GP = 7;
        break;
    case 'D':
        GP = 6;
        break;
    case 'E':
        GP = 5;
        break;
    case 'P':
        GP = 4;
        break;
    }

    semester_TGC = semester_TGC + (GP * credit);
    semester_TC = semester_TC + credit;
}

SGPA = (float)semester_TGC / semester_TC;
printf("\nSemester %d SGPA = %.2f\n", semester, SGPA);
cumulative_TGC = cumulative_TGC + semester_TGC;
cumulative_TC = cumulative_TC + semester_TC;
}

CGPA = (float)cumulative_TGC / cumulative_TC;
printf("\n----- Final Result ----- \n");
printf("Global ID = %d\n", global_id);
printf("Number of Semesters = %d\n", m);
printf("Cumulative Total Credits = %d\n", cumulative_TC);
printf("CGPA = %.2f\n", CGPA);
return 0;

```

}

4. Compilation and Execution

```
(kali㉿kali)-[~]
$ vi hdoc19.c

(kali㉿kali)-[~]
$ gcc hdoc19.c -o hdoc19

(kali㉿kali)-[~]
$ ./hdoc19
Enter Global ID: 101
Enter number of semesters: 1

Semester 1
Enter number of courses: 2

Course 1
Enter course code: 1001
Enter course credit: 4
Enter grade (O/A/B/C/D/E/P): O

Course 2
Enter course code: 1002
Enter course credit: 3
Enter grade (O/A/B/C/D/E/P): A

Semester 1 SGPA = 9.57

—— Final Result ——
Global ID = 101
Number of Semesters = 1
```

```
Cumulative Total Credits = 7
CGPA = 9.57

(kali㉿kali)-[~]
$ gcc hdoc19.c -o hdoc19

(kali㉿kali)-[~]
$ ./hdoc19
Enter Global ID: 102
Enter number of semesters: 2

Semester 1
Enter number of courses: 2

Course 1
Enter course code: 1001
Enter course credit: 3
Enter grade (O/A/B/C/D/E/P): A

Course 2
Enter course code: 1002
Enter course credit: 4
Enter grade (O/A/B/C/D/E/P): B

Semester 1 SGPA = 8.43

Semester 2
Enter number of courses: 2
```

```
Course 1
Enter course code: 2001
Enter course credit: 3
Enter grade (O/A/B/C/D/E/P): 0
```

```
Course 2
Enter course code: 2002
Enter course credit: 2
Enter grade (O/A/B/C/D/E/P): C
```

```
Semester 2 SGPA = 8.80
```

```
—— Final Result ——
Global ID = 102
Number of Semesters = 2
Cumulative Total Credits = 12
CGPA = 8.58
```

```
(kali㉿kali)-[~]
$ gcc hdoc19.c -o hdoc19
```

```
(kali㉿kali)-[~]
$ ./hdoc19
Enter Global ID: 0
Invalid Global ID. Please enter a positive value.
Enter Global ID: 103
Enter number of semesters: -2
Invalid number of semesters. Please enter a positive value.
Enter number of semesters: 1
```

```
Semester 1
Enter number of courses: 0
Invalid number of courses. Please enter a positive value.
Enter number of courses: 1
```

```
Course 1
Enter course code: -10
Invalid course code. Please enter a positive value.
Enter course code: 1001
Enter course credit: 0
Invalid credit. Please enter a positive value.
Enter course credit: 3
Enter grade (O/A/B/C/D/E/P): X
Invalid grade. Please enter O, A, B, C, D, E, or P.
Enter grade (O/A/B/C/D/E/P): B
```

```
Semester 1 SGPA = 8.00
```

```
—— Final Result ——
Global ID = 103
Number of Semesters = 1
Cumulative Total Credits = 3
CGPA = 8.00
```

```
(kali㉿kali)-[~]
$ gcc hdoc19.c -o hdoc19
```

```
(kali㉿kali)-[~]
$ ./hdoc19
```

```
Enter Global ID: 1
Enter number of semesters: 1

Semester 1
Enter number of courses: 1

Course 1
Enter course code: 1
Enter course credit: 1
Enter grade (O/A/B/C/D/E/P): P
```

Semester 1 SGPA = 4.00

```
—— Final Result ——
Global ID = 1
Number of Semesters = 1
Cumulative Total Credits = 1
CGPA = 4.00
```

```
(kali@kali)-[~]
$ gcc hdoc19.c -o hdoc19
```

```
(kali@kali)-[~]
$ ./hdoc19
Enter Global ID: 1
Enter number of semesters: 1
```

```
Semester 1
Enter number of courses: 1
```

```
Course 1
Enter course code: 1
Enter course credit: 1
Enter grade (O/A/B/C/D/E/P): O
```

Semester 1 SGPA = 10.00

```
—— Final Result ——
Global ID = 1
Number of Semesters = 1
Cumulative Total Credits = 1
CGPA = 10.00
```